

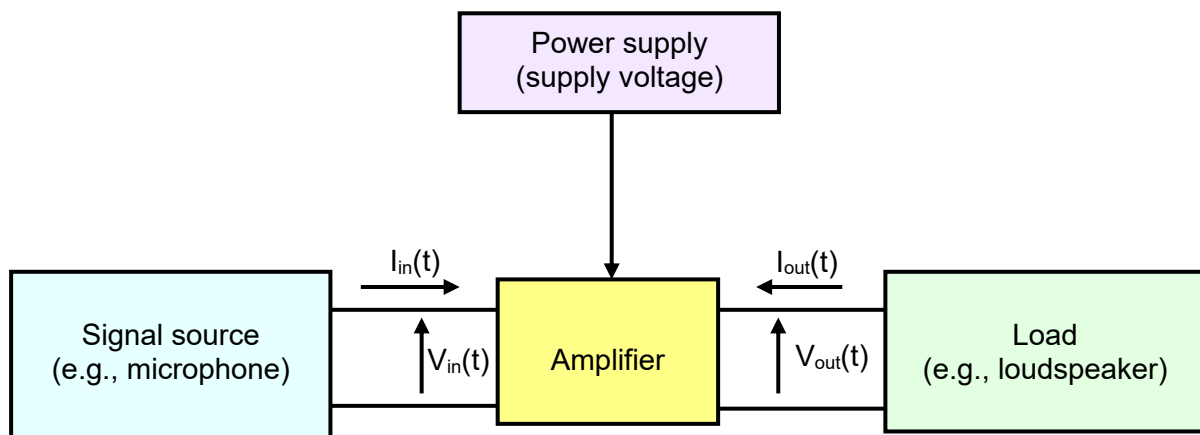
Linear Amplifiers

S Le Goff

An analogue circuit that acts as an amplifier reproduces changes in its input signal as proportionately larger changes in its output signal. The term signal is used to denote the information-carrying fluctuations of a given voltage or current.

The amplification function performed by the circuit can be either linear or non-linear. If the amplification is linear, the output signal will be an amplified replica of the input signal, thus implying that no distortion will be introduced during the amplification process. If the amplification is non-linear, the output signal will be correlated to the input signal, but not an exact replica of it.

Here, we are only concerned with linear amplifiers for which the output signal is, at least in theory, proportional to the input signal. The information-carrying signal to be amplified can be either a voltage or a current.



Amplifiers are active circuits (unlike resistors, capacitors and diodes) since the output signal magnitude is greater than the input signal magnitude. Such a result can be obtained only when using active components such as transistors. The amplifier requires some form of power supply to enable it to boost the input signal.

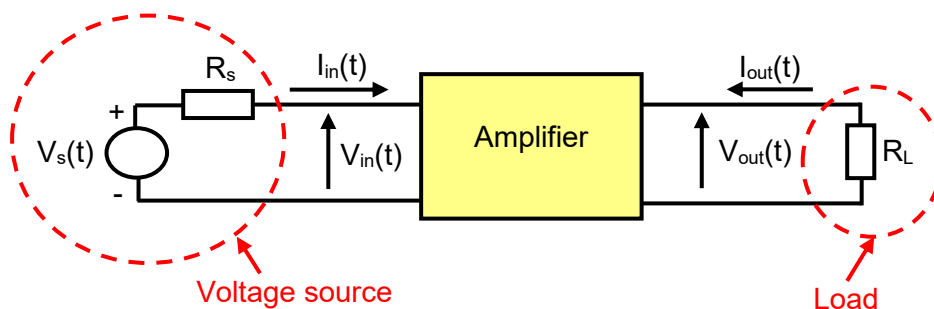
In audio systems, the source of the input signal can be a CD or DVD player, the microphone in a telephone handset, the antenna of a radio receiver, etc. The related voltage signal can

range from a few μV to a few mV , and the current can range from a few μA to a few mA . This corresponds to a very low level of electric power, not sufficient to drive many loads. For example, a system of speakers requires a power that can reach in some cases hundreds or thousands of watts.

The issue of voltage and current amplification concerns a huge number of signal sources that are very different from each other, e.g. a fuel-level sensor in an automobile gas tank or a smoke detector in a fire alarm system.

There are four different ways to model a given amplifier circuit. The model we must select depends on the application for which the circuit is intended to be used.

For many applications, the information source generates a time-varying voltage. When the load circuit needs to be driven by an amplified replica of this voltage, we must employ a circuit that performs well as a *voltage amplifier* and it makes sense to also model this circuit as a voltage amplifier. However, if the load must be driven by a current rather than a voltage, we become more interested in the performance of the same circuit as a *trans-conductance amplifier*. It is then preferable to model this circuit as a trans-conductance amplifier.



With a voltage-driven load, we are interested in the (real) voltage gain defined as the ratio $V_{out}(t)/V_s(t)$.

With a current-driven load, we are interested in the (real) trans-conductance gain defined as the ratio $I_{out}(t)/V_s(t)$

In the diagram shown above, the quantities $V_{out}(t)$ and $I_{out}(t)$ represent the voltage and current transferred to the load circuit, respectively, while the quantities $V_{in}(t)$ and $I_{in}(t)$ represent the voltage and current available at the circuit input, respectively.

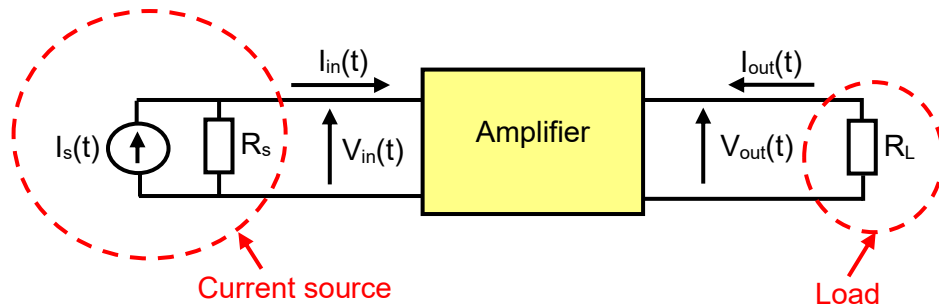
Note that the quantity $V_{in}(t)$ is NOT the voltage generated by the source, i.e. $V_{in}(t) \neq V_s(t)$. In other words, focusing on the voltage gain issue as an example, it would be a mistake to believe that the *real gain* of a voltage amplifier is given by the ratio $V_{out}(t)/V_{in}(t)$. Instead, the real voltage gain of the system must be defined as the ratio $V_{out}(t)/V_s(t)$. Ultimately, what truly matters to the user is the latter ratio, and not the ratio $V_{out}(t)/V_{in}(t)$.

Examples of applications where a voltage or trans-conductance amplifier is required:

- (1) An electrocardiogram (ECG) is used to detect the heart's electrical activity to identify eventual heart disorders. It utilises a pair of electrodes to reveal the ionic potential difference between their respective points of application on the skin. The voltage detected between the electrodes is around 1 – 5 mV.
- (2) A thermocouple produces a voltage proportional to a temperature.
- (3) A light-to-voltage (LTV) sensor provides a voltage output proportional to light.
- (4) Most microphones produce a voltage signal from mechanical vibrations.
- (5) Bio-signals are recorded as very low-level voltages (from 1 μ V to 100 mV) generated by nerves, muscles, etc.
- (6) The brain waves use electroencephalography (EEG) to furnish potential differences in the microvolt range measured across locations on the user's scalp.

For these examples, voltage amplification is typically required for further signal processing or to simply drive a display device or a loudspeaker system. However, trans-conductance amplification is typically required when the amplified signal is used to drive a meter.

There are also applications for which the information source generates a time-varying current. When the load circuit needs to be driven by an amplified replica of this current, we must employ a circuit that performs well as a *current amplifier* and also model this circuit as a current amplifier. Now, if the load must be driven by a voltage rather than a current, we are more interested in the performance of the same circuit as a *trans-resistance amplifier* and it is then preferable to model this circuit as a trans-resistance amplifier.



With a current-driven load, we are interested in the (real) current gain defined as the ratio $I_{out}(t)/I_s(t)$.
 With a voltage-driven load, we are interested in the (real) trans-resistance gain defined as the ratio $V_{out}(t)/I_s(t)$

It is worth noting that the quantity $I_{in}(t)$ is NOT the current generated by the source, i.e. $I_{in}(t) \neq I_s(t)$. In other words, focusing on the current gain issue as an illustration, it would be a mistake to believe that the *real gain* of a current amplifier is given by the ratio $I_{out}(t)/I_{in}(t)$. Instead, the real current gain of the system must be defined as the ratio $I_{out}(t)/I_s(t)$. Ultimately, what truly matters to the user is the latter ratio, and not the ratio $I_{out}(t)/I_{in}(t)$.

Example of applications where a current or trans-resistance amplifier is required: Photovoltaic cells and photodiodes act like current sources since they generate a current proportional to the intensity of light. Trans-resistance amplification is typically required for further signal processing or to simply drive a display device or a loudspeaker system. However, current amplification is typically required when the amplified signal is used to drive a meter.

We must be aware that some circuits perform very well as, say, both current and voltage amplifiers, while some others are only suitable for amplifying either a voltage or a current. The choice of a particular circuit will therefore entirely depend on its application.

In the remainder of this chapter, we will focus our attention on the voltage and current amplifier models, and mostly ignore the trans-conductance and trans-resistance amplifier models. There is indeed no need to repeat explanations that are very similar for the four models. The extrapolation of the results obtained for the voltage and current amplifier models to the trans-conductance and trans-resistance amplifier models is actually straightforward and thus left as a (probably boring) exercise to the reader.

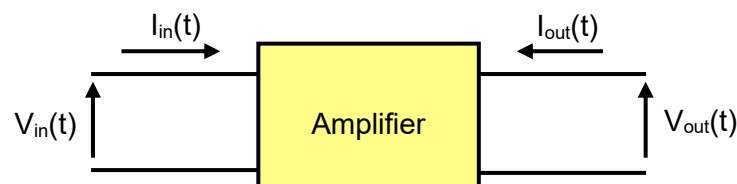
• Equivalent Model of an Amplifier

In order for an amplifier to perform some useful function, a circuit (called the *source*) must be connected to its input and another circuit (called the *load*) must be connected to its output. The source circuit provides the amplifier with the signal (voltage or current) to be amplified, whereas the load circuit makes use of the signal (voltage or current) produced by the amplifier.

An ideal amplifier would always give an output signal that is determined only by the input signal and gain, irrespective of what is connected to its output. Also, an ideal amplifier would not affect the signal produced by the source circuit. In fact, real amplifiers do not fulfil these requirements.

This is why we need to introduce in this chapter the concepts of input and output resistances of an amplifier. We will see that the values of these resistances can strongly affect the overall gain when several circuits/amplifiers are connected to each other.

Any amplifier, no matter how complicated it is, can be considered by the outside world as a two-port device: It can be viewed as a black box with an input port as well as an output port.



To model each port, we can simply use the Thévenin or Norton theorems, depending on whether we choose to view our circuit as either a *voltage amplifier* (Thévenin model for both ports) or a *current amplifier* (Norton model for both ports) or a *trans-conductance amplifier* (Thévenin model for the input port and Norton model for the output port), or a *trans-impedance amplifier* (Norton model for the input port and Thévenin model for the output port).

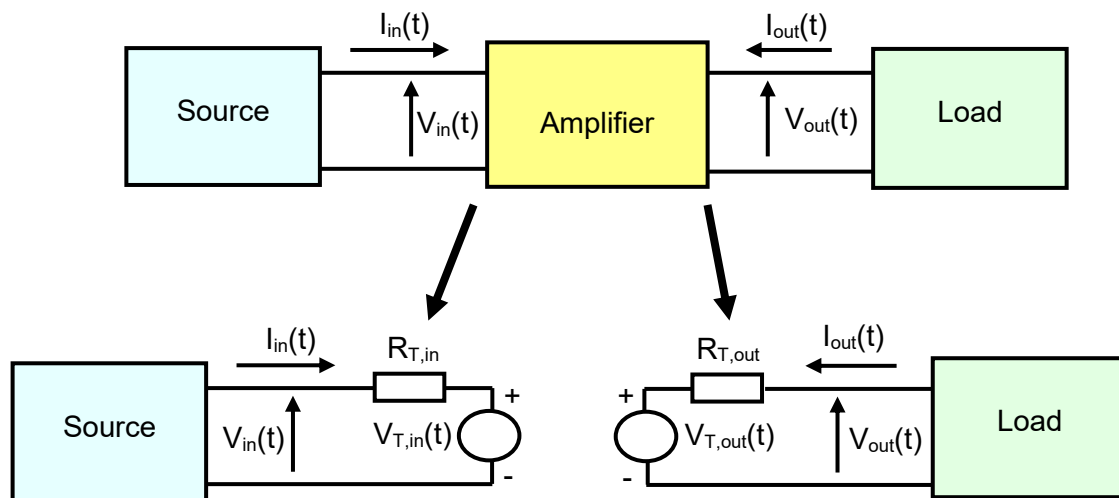
The Thévenin theorem, derived by the French electrical engineer Léon Charles Thévenin, holds that any linear electric circuit with voltage and current sources as well

as resistances can be replaced by an equivalent voltage source V_T connected in series with an equivalent resistance R_T .

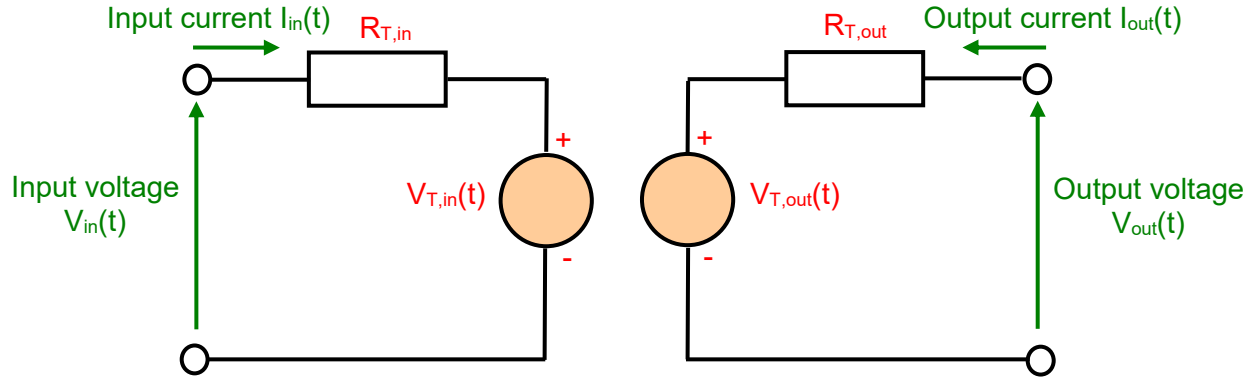
The Norton theorem, derived by the American electrical engineer Edward Lawry Norton, holds that any linear electric circuit with voltage and current sources as well as resistances can be replaced by an equivalent current source I_N connected in parallel with an equivalent resistance R_N .

• Voltage Amplifier Model

Thévenin theorem states that the source circuit “sees” the amplifier as a voltage source $V_{T,in}(t)$ in series with a resistance $R_{T,in}$, whereas the load circuit “sees” the amplifier as a voltage source $V_{T,out}(t)$ in series with a resistance $R_{T,out}$.

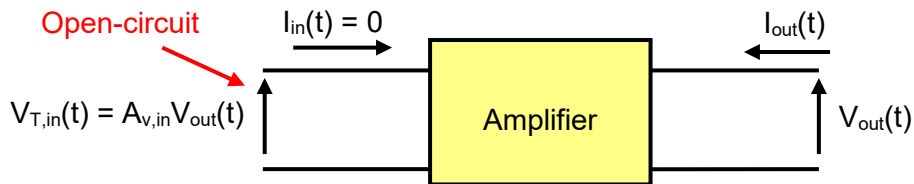


Hence, the corresponding model of a voltage amplifier is as shown below.



Based on this model, we can write the following equations: $V_{in}(t) = R_{T,in} \cdot I_{in}(t) + V_{T,in}(t)$ and $V_{out}(t) = R_{T,out} \cdot I_{out}(t) + V_{T,out}(t)$. From these two equations, it appears that the values of the four parameters $R_{T,in}$, $R_{T,out}$, $V_{T,in}(t)$, and $V_{T,out}(t)$ can be determined as explained below.

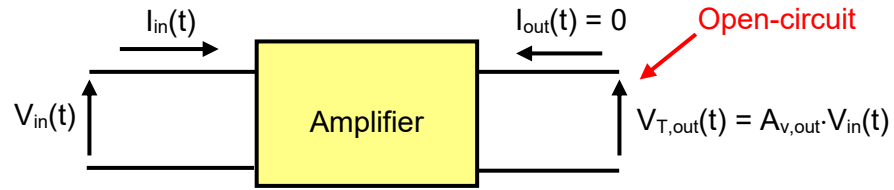
- $V_{T,in}(t)$ is the voltage at the amplifier input when $I_{in}(t) = 0$ (open-circuit between input terminals).



We will see that $V_{T,in}(t) = A_{v,in} \cdot V_{out}(t)$, where $A_{v,in}$ designates the *open-circuit (OC) reverse voltage gain* of the amplifier. The term “open-circuit” refers to the fact that this voltage gain is obtained when the amplifier input is not connected to any source circuit, i.e. when there is an open-circuit at the input. The adjective “reverse” refers to the fact that the quantity $V_{T,in}(t)$ depends on the output voltage $V_{out}(t)$, which indicates that the amplifier operates in the reverse direction. We can finally write

$$A_{v,in} = \left. \frac{V_{in}(t)}{V_{out}(t)} \right|_{I_{in}(t)=0} .$$

- $V_{T,out}(t)$ is the voltage at the amplifier output when $I_{out}(t) = 0$ (open-circuit between output terminals).

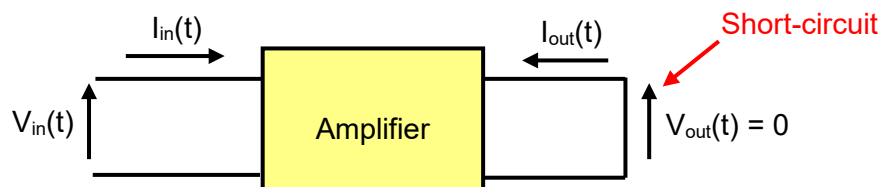


We will see that $V_{T,out}(t) = A_{v,out} \cdot V_{in}(t)$, where $A_{v,out}$ designates the *open-circuit (OC) forward voltage gain* of the amplifier. The term “open-circuit” refers to the fact that this voltage gain is obtained when the amplifier output is not connected to any load circuit, i.e. when there is an open-circuit between the output terminals. The adjective “forward” refers to the fact that the quantity $V_{T,out}(t)$ depends on the input voltage $V_{in}(t)$, which indicates that the amplifier operates in the forward direction. We can finally write

$$A_{v,out} = \left. \frac{V_{out}(t)}{V_{in}(t)} \right|_{I_{out}(t)=0}.$$

- The resistance $R_{T,in}$ is the ratio between $V_{in}(t)$ and $I_{in}(t)$ when $V_{T,in}(t) = 0$, i.e. when $V_{out}(t) = 0$ (short circuit between output terminals):

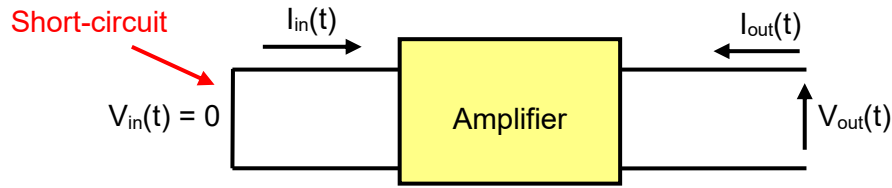
$$R_{T,in} = \left. \frac{V_{in}(t)}{I_{in}(t)} \right|_{V_{out}(t)=0}.$$



The quantity $R_{T,in}$ is referred to as the *input resistance* of the amplifier and hereafter simply denoted as R_{in} .

- The resistance $R_{T,out}$ is the ratio between $V_{out}(t)$ and $I_{out}(t)$ when $V_{T,out}(t) = 0$, i.e. when $V_{in}(t) = 0$ (short circuit between input terminals):

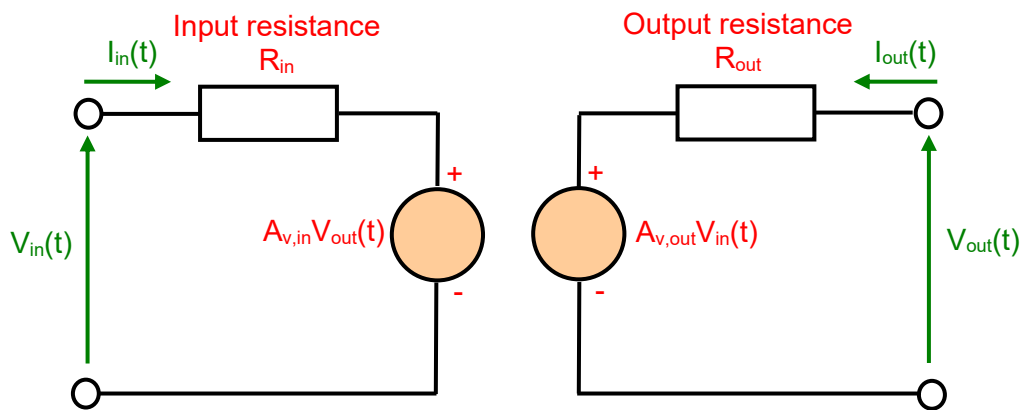
$$R_{T,out} = \left. \frac{V_{out}(t)}{I_{out}(t)} \right|_{V_{in}(t)=0}.$$



The quantity $R_{T,out}$ is referred to as the *output resistance* of the amplifier and hereafter simply denoted as R_{out} .

The voltage amplifier model thus obtained will be hereafter referred to as *bilateral voltage amplifier model*.

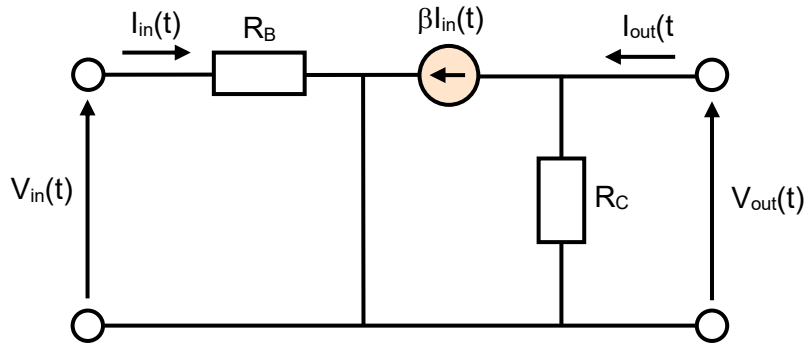
Bilateral voltage amplifier model



Bilateral voltage amplifier model parameter	How to determine its expression/value?
$A_{v,in}$ - OC reverse voltage gain	$V_{in}(t)/V_{out}(t)$ with OC @ input ($I_{in}(t) = 0$)
$A_{v,out}$ - OC forward voltage gain	$V_{out}(t)/V_{in}(t)$ with OC @ output ($I_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$ with SC* @ output ($V_{out}(t) = 0$)
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with SC @ input ($V_{in}(t) = 0$)

* short circuit

• **Example:** Let us determine the voltage amplifier model for the circuit depicted below. Note that this circuit corresponds to a generic *common-emitter* amplifier circuit.



The general equations for this circuit are as follows:

$$(1) V_{in}(t) = R_B I_{in}(t) \text{ and } (2) V_{out}(t) = R_C (I_{out}(t) - \beta I_{in}(t)).$$

- OC reverse voltage gain: $I_{in}(t) = 0 \Rightarrow V_{in}(t) = 0 \Rightarrow A_{v,in} = \left. \frac{V_{in}(t)}{V_{out}(t)} \right|_{I_{in}(t)=0} = 0.$

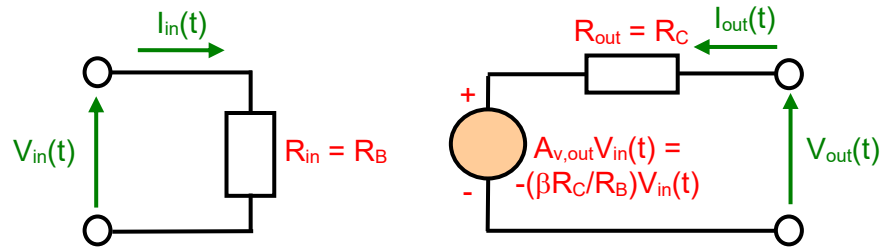
- OC forward voltage gain: $I_{out}(t) = 0 \Rightarrow V_{out}(t) = -R_C \beta I_{in}(t) = -\beta \frac{R_C}{R_B} V_{in}(t)$

$$\Rightarrow A_{v,out} = \left. \frac{V_{out}(t)}{V_{in}(t)} \right|_{I_{out}(t)=0} = -\beta \frac{R_C}{R_B}.$$

- Input resistance: $R_{in} = \left. \frac{V_{in}(t)}{I_{in}(t)} \right|_{V_{out}(t)=0} = R_B.$

- Output resistance: $R_{out} = \left. \frac{V_{out}(t)}{I_{out}(t)} \right|_{V_{in}(t)=0} = R_C.$

The circuit considered in this example can thus be viewed as a voltage amplifier whose model is depicted below.



• The Unilateral Voltage Amplifier Model as a Simplification of the Bilateral Voltage Amplifier Model

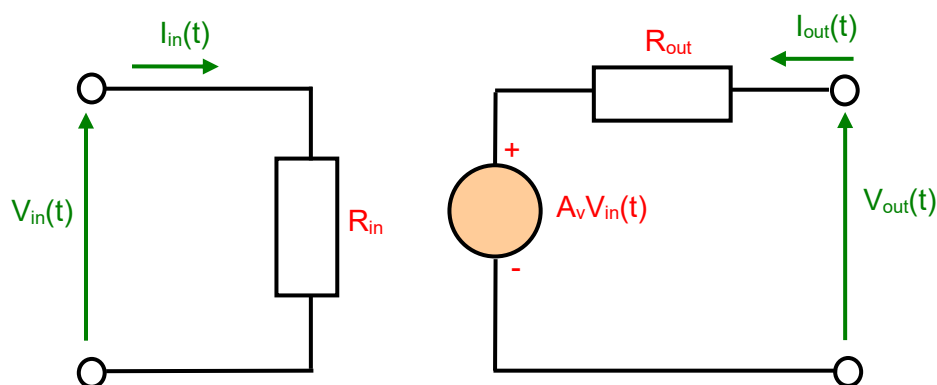
The problem with the bilateral voltage amplifier model is that it is too complex to handle when inserted into a larger circuit, essentially because there are too many (four) parameters in it.

Fortunately, for *most* circuits, the bilateral model can be simplified because the OC reverse voltage gain $A_{v,in}$ for these circuits is equal to zero.

In order to make our journey through this chapter more comfortable, we are going to assume, from now on, that $A_{v,in} = 0$. We will need to bear in mind that this assumption does not hold for a few important circuits such as the common-collector amplifier.

Under the assumption $A_{v,in} = 0$, an amplifier circuit can be viewed as a voltage amplifier and replaced inside a larger circuit by a model defined by the three parameters A_v , R_{in} , and R_{out} . This model is known as the *unilateral voltage amplifier model*.

Unilateral voltage amplifier model



In this unilateral model, the OC forward voltage gain $A_{v,out}$ is referred to as *OC voltage gain*, since there is now no need to make a distinction between a reverse and a forward gain, and simply denoted as A_v .

The quantity A_v is the voltage gain across the two output terminals when the load of the amplifier is an open circuit ($I_{out}(t) = 0$). This is, in other words, the voltage gain of the amplifier in isolation.

As mentioned earlier, the quantity A_v is referred to as *open-circuit (OC) voltage gain* because it is the ratio of the output voltage to the input voltage in the absence of any loading effects. We can write

$$A_v = \left. \frac{V_{out}(t)}{V_{in}(t)} \right|_{I_{out}(t)=0}.$$

The input resistance R_{in} is defined as $R_{in} = \frac{V_{in}(t)}{I_{in}(t)}$.

Since $A_{v,in}$ is equal to zero, the condition $V_{out}(t) = 0$ does no longer have to be satisfied when evaluating the input resistance.

The output resistance R_{out} is defined as $R_{out} = \left. \frac{V_{out}(t)}{I_{out}(t)} \right|_{V_{in}(t)=0}$.

Unilateral voltage amplifier model parameter	How to determine its expression/value?
A_v - OC voltage gain	$V_{out}(t)/V_{in}(t)$ with OC @ output ($I_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with SC @ input ($V_{in}(t) = 0$)

As a summary, the unilateral voltage amplifier model is applicable to all circuits for which $A_{v,in} = 0$ (e.g., common-emitter, common-base, common-source, common-drain, and common-gate amplifiers). Due to its simplicity, this model is rather easy to handle in a larger circuit. There are however a few circuits (e.g., common-collector and Darlington-pair amplifiers) for which $A_{v,in} \neq 0$. Hence, these circuits do not have a unilateral voltage amplifier model, and the

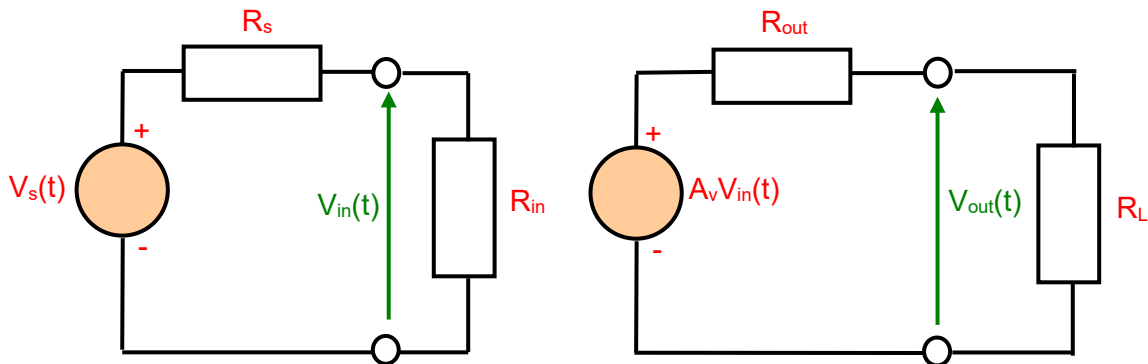
only way to view them in a larger circuit is to employ the (very complex) bilateral voltage amplifier model.

• Real Voltage Gain

For simplicity's sake, we only consider hereafter the case of a unilateral voltage amplifier model.

The OC voltage gain A_v is not the real voltage gain obtained when an amplifier is connected to a source circuit and a load circuit. In fact, the real voltage gain, hereafter denoted as $\tilde{A}_v$, depends not only on A_v but also on the other parameters of the circuit including both resistances R_{in} and R_{out} .

To illustrate this statement, let us consider the generic circuit depicted below. The input of the amplifier is connected to a voltage source $V_s(t)$ with an output resistance R_s . The load resistance R_L , that represents the input resistance of the load circuit, is connected to the output of the amplifier.



We can show that the voltage gain of the whole circuit is given by

$$\tilde{A}_v = \frac{V_{out}(t)}{V_s(t)} = A_v \frac{R_{in}}{R_{in} + R_s} \frac{R_L}{R_L + R_{out}}.$$

This expression shows that, to maximise the overall gain $\tilde{A}_v$, one must ensure that $R_{in} \gg R_s$ and $R_{out} \ll R_L$. In other words, to maximise the transfer of voltage from source to load, we must ensure that both conditions $R_{in} \gg R_s$ and $R_{out} \ll R_L$ are satisfied.

In order to obtain $\tilde{A}_v = A_v$, we would ideally need to have $R_{in} = +\infty$ and $R_{out} = 0$. These are the characteristics of an ideal voltage amplifier.

As a summary, a “good” voltage amplifier circuit has a large input resistance, a low output resistance, and a large OC voltage gain.

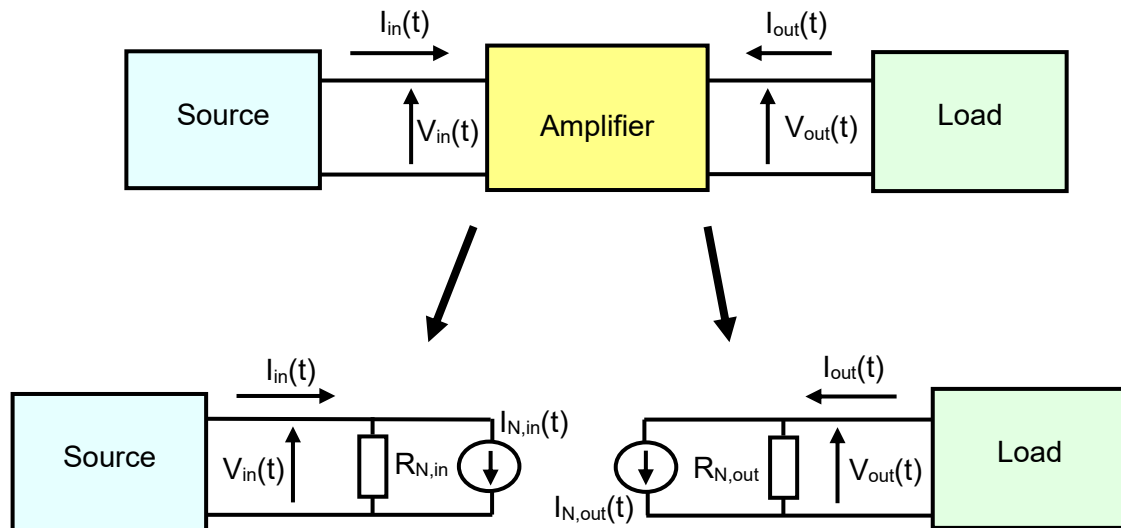
This example shows that, when an amplifier is connected to a source and a load, the resulting output voltage may be considerably less than one might have expected given the source voltage and the OC voltage gain of this amplifier. This is a phenomenon known as the *loading effect*.

The actual input voltage $V_{in}(t)$ to the amplifier is less than the source voltage $V_s(t)$ due to the effects of the voltage divider formed by R_s and R_{in} . Similarly, the output voltage of the circuit is affected by the load resistance R_L due to the effects of the voltage divider formed by R_L and R_{out} .

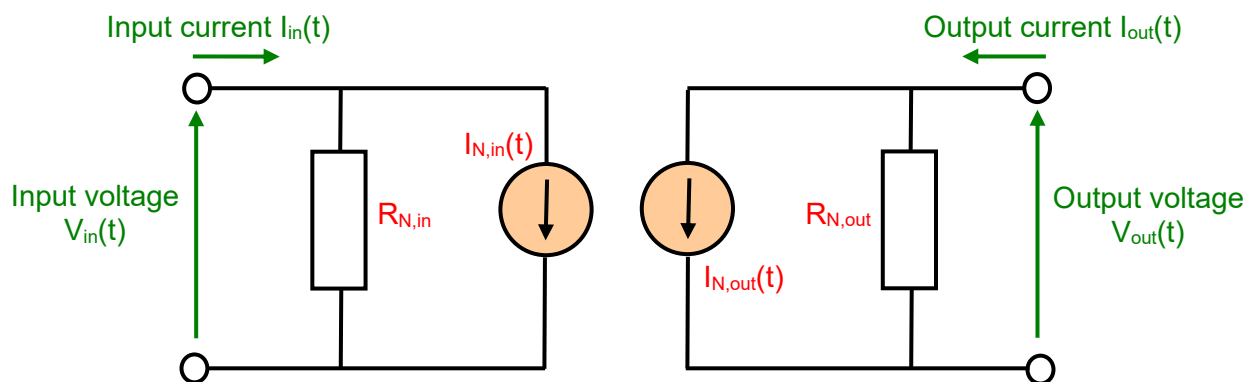
• Current Amplifier

What to do in case we are more interested in current gains than voltage gains? The answer is that we should view our circuit as a current amplifier rather than a voltage amplifier. This is done by applying Norton’s theorem at both input and output ports of our circuit.

Norton theorem states that the source circuit “sees” the amplifier as a current source $I_{N,in}(t)$ in parallel with a resistance $R_{N,in}$, whereas the load circuit “sees” the amplifier as a current source $I_{N,out}(t)$ in parallel with a resistance $R_{N,out}$.

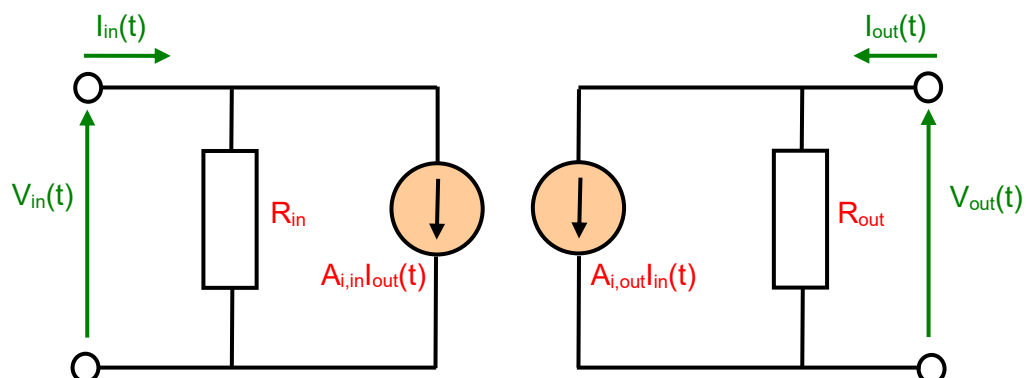


Therefore, the current amplifier model of a circuit is as shown below.



Based on an analysis very similar to that performed for the voltage amplifier, we could show that this model is in fact equivalent to that depicted below.

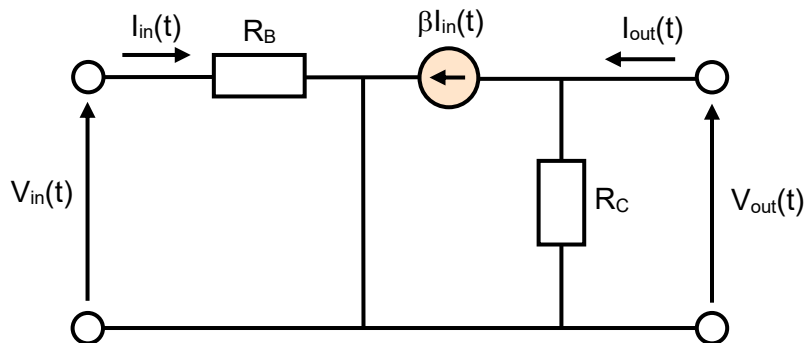
Bilateral current amplifier model



The bilateral current amplifier model is completely defined by four parameters which can be determined as indicated in the table below.

Bilateral current amplifier model parameter	How to determine its expression/value?
$A_{i,in}$ - SC reverse current gain	$I_{in}(t)/I_{out}(t)$ with SC @ input ($V_{in}(t) = 0$)
$A_{i,out}$ - SC forward current gain	$I_{out}(t)/I_{in}(t)$ with SC @ output ($V_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$ with OC @ output ($I_{out}(t) = 0$)
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with OC @ input ($I_{in}(t) = 0$)

• **Example:** Let us determine the current amplifier model for the circuit depicted below. This circuit corresponds to the generic common-emitter amplifier circuit already considered previously.



The general equations for this circuit are (of course) identical to those previously obtained:

(1) $V_{in}(t) = R_B I_{in}(t)$ and (2) $V_{out}(t) = R_C (I_{out}(t) - \beta I_{in}(t))$.

• SC reverse current gain: $V_{in}(t) = 0 \Rightarrow I_{in}(t) = V_{in}(t)/R_B = 0$

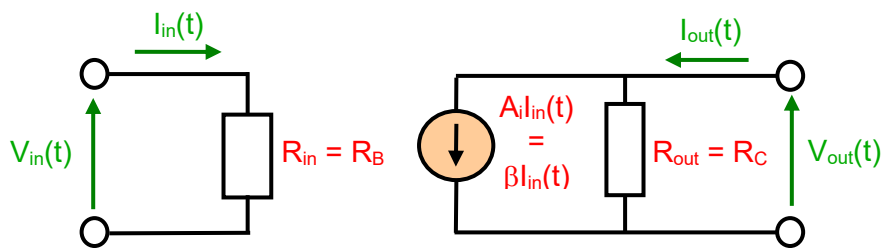
$$\Rightarrow A_{i,in} = \left. \frac{I_{in}(t)}{I_{out}(t)} \right|_{V_{in}(t)=0} = 0.$$

• SC forward current gain: $V_{out}(t) = 0 \Rightarrow I_{out}(t) = \beta I_{in}(t) \Rightarrow A_{i,out} = \left. \frac{I_{out}(t)}{I_{in}(t)} \right|_{V_{out}(t)=0} = \beta.$

- Input resistance: $R_{in} = \left. \frac{V_{in}(t)}{I_{in}(t)} \right|_{I_{out}(t)=0} = R_B$.

- Output resistance: $R_{out} = \left. \frac{V_{out}(t)}{I_{out}(t)} \right|_{I_{in}(t)=0} = R_C$.

As a conclusion, the current amplifier model of the circuit considered in this example is shown below.



• The Unilateral Current Amplifier Model as a Simplification of the Bilateral Current Amplifier Model

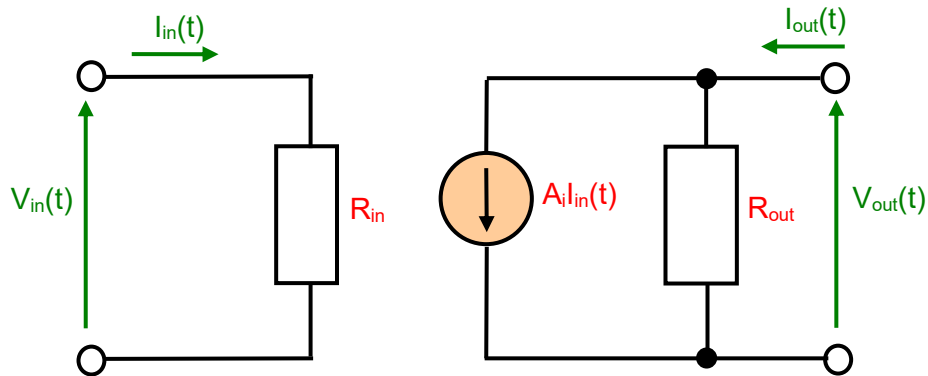
Due to the complexity of the bilateral current amplifier model, it is preferable to replace it by a simplified version that will be much easier to handle. To do so, we can assume that the SC reverse current gain $A_{i,in}$ is equal to zero, which is indeed the case with most circuits.

Hence, from now on, we are going to assume that $A_{i,in} = 0$, but we must always remember to be careful because this assumption is not valid for a few important circuits.

Under this assumption, any amplifier circuit can be viewed as a current amplifier, i.e. replaced inside a larger circuit by a model defined by the three parameters A_i , R_{in} , and R_{out} . This model is known as the *unilateral current amplifier model*.

Note that the notation $A_{i,out}$ has now been replaced by the notation A_i that simply denotes the SC current gain (as there is no need to make a distinction between reverse and forward current gains anymore).

Unilateral current amplifier model



The quantity A_i is the current gain across the two output terminals when the load of the amplifier is a short circuit ($V_{out}(t) = 0$). This is, in other words, the current gain of the amplifier in isolation. The quantity A_i is referred to as *short-circuit current gain* because it is the ratio of the output current to the input current in the absence of any loading effects. We can write

$$A_i = \left. \frac{I_{out}(t)}{I_{in}(t)} \right|_{V_{out}(t)=0}$$

The input resistance R_{in} is defined as $R_{in} = \frac{V_{in}(t)}{I_{in}(t)}$.

Note that, since $A_{i,in} = 0$, the condition $I_{out}(t) = 0$ does not have to be satisfied when evaluating the input resistance.

The output resistance R_{out} is defined as $R_{out} = \left. \frac{V_{out}(t)}{I_{out}(t)} \right|_{I_{in}(t)=0}$.

Unilateral current amplifier model parameter	How to determine its expression/value?
A_i - SC current gain	$I_{out}(t)/I_{in}(t)$ with SC @ output ($V_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with OC @ input ($I_{in}(t) = 0$)

As a summary, the unilateral current amplifier model is applicable to all circuits for which $A_{i,in} = 0$ (e.g., common-emitter, common-base, common-source, common-drain, and common-gate amplifiers). Due to its simplicity, this model is rather easy to handle in a larger circuit.

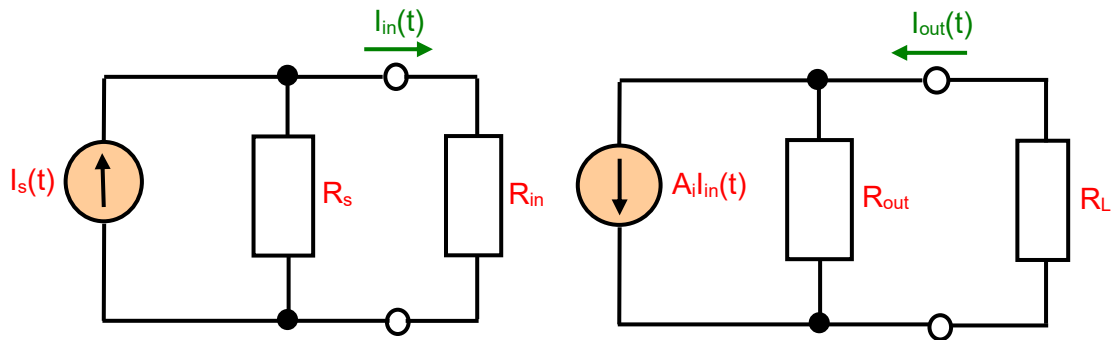
There are however a few circuits (e.g., common-collector and Darlington-pair amplifiers) for which $A_{i,in} \neq 0$. Hence, these circuits do not have a unilateral current amplifier model, and the only way to view them in a larger circuit is to employ the (very complex) bilateral current amplifier model.

• Real Current Gain

For simplicity sake, we only consider hereafter the case of a unilateral current amplifier model.

In practice, the SC current gain A_i is not the real current gain obtained when an amplifier is connected to a source circuit and a load circuit. In fact, this real current gain, hereafter denoted as $\tilde{A}_i$, depends not only on A_i but also on the other parameters of the circuit including, of course, the resistances R_{in} and R_{out} .

To illustrate this statement, let us consider the circuit depicted below. The input of an amplifier is connected to a current source $I_s(t)$ with an output resistance R_s . The load resistance R_L , that represents the input resistance of the loading circuit, is connected to the output of the amplifier.



We can easily show that the current gain of the whole circuit is given by

$$\tilde{A}_i = \frac{I_{out}(t)}{I_s(t)} = A_i \frac{R_s}{R_{in} + R_s} \frac{R_{out}}{R_L + R_{out}}.$$

This expression shows that, to maximise the overall current gain $\tilde{A}_i$, it is important to make sure that R_{in} is as small as possible (i.e., $R_{in} \ll R_s$) and R_{out} is as large as possible (i.e., $R_{out} \gg R_L$). In other words, to maximise the transfer of current from source to load, we must ensure that both conditions $R_{in} \ll R_s$ and $R_{out} \gg R_L$ are satisfied.

In order to obtain $\tilde{A}_i = A_i$, we would ideally need to have $R_{in} = 0$ and $R_{out} = +\infty$. These are the characteristics of an ideal current amplifier.

A “good” current amplifier has a small input resistance, a large output resistance, and a large SC current gain.

• Characteristics of a “Good” Amplifier

We have explained how to determine the voltage and current amplifier models for any electronic circuit. We could also study in a very similar fashion the techniques to find the trans-conductance and trans-resistance amplifier models.

However, that would be very repetitive and boring. Instead, we simply summarise in the two tables shown below the results that would be obtained for both models.

Only the unilateral models are considered here for simplicity purposes.

Unilateral trans-conductance amplifier model parameter	How to determine its expression/value?
A_g - SC trans-conductance gain (expressed in siemens, S)	$I_{out}(t)/V_{in}(t)$ with SC @ output ($V_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with SC @ input ($V_{in}(t) = 0$)

Unilateral trans-resistance amplifier model parameter	How to determine its expression/value?
A_r - OC trans-resistance gain (expressed in ohms, Ω)	$V_{out}(t)/I_{in}(t)$ with OC @ output ($I_{out}(t) = 0$)
R_{in} – input resistance	$V_{in}(t)/I_{in}(t)$
R_{out} – output resistance	$V_{out}(t)/I_{out}(t)$ with OC @ input ($I_{in}(t) = 0$)

Finally, the table shown below summarises the characteristics of a “good” amplifier depending on the adopted model.

Simplified amplifier model	Characteristics of a “good” amplifier
Voltage amplifier	High OC voltage gain A_v Large input resistance, low output resistance
Current amplifier	High SC current gain A_i Low input resistance, large output resistance
Trans-conductance amplifier	High SC trans-conductance gain A_g Large input resistance, large output resistance
Trans-resistance amplifier	High OC trans-resistance gain A_r Low input resistance, low output resistance

• Power Amplification

All circuits considered throughout these lecture notes can amplify a voltage and/or a current. This implies that they can also amplify the signal power. The power available from the source is given by $P_s(t) = V_s(t) \cdot I_s(t)$, whereas the power delivered to the load is given by $P_{out}(t) = V_{out}(t) \cdot I_{out}(t)$.

The power gain is defined as the ratio of the power supplied by the amplifier to the load to the power available from the source:

$$A_p = \frac{P_{out}(t)}{P_s(t)} = \frac{V_{out}(t) \cdot I_{out}(t)}{V_s(t) \cdot I_s(t)} = \tilde{A}_v \cdot \tilde{A}_i.$$

We previously showed that, to maximise the transfer of voltage from source to load, we must ensure that both conditions $R_{in} \gg R_s$ and $R_{out} \ll R_L$ are satisfied. However, we also showed that, to maximise the transfer of current from source to load, we must ensure that both conditions $R_{in} \ll R_s$ and $R_{out} \gg R_L$ are satisfied.

These two conflicting sets of conditions clearly indicate that it is impossible to maximise the transfer of voltage and the transfer of current at the same time.

For some applications, we are actually required to maximise the transfer of electric power from source to load. Under what conditions can this be achieved?

We have already shown that the power gain is given by

$$A_p = \tilde{A}_v \cdot \tilde{A}_i = A_v \frac{R_{in}}{R_{in} + R_s} \frac{R_L}{R_L + R_{out}} \cdot A_i \frac{R_s}{R_{in} + R_s} \frac{R_{out}}{R_L + R_{out}},$$

which can also be written as

$$A_p = A_v A_i \frac{R_{in} R_s}{(R_{in} + R_s)^2} \frac{R_L R_{out}}{(R_L + R_{out})^2},$$

where A_v and A_i designate the OC voltage and SC current gains of this amplifier, respectively.

The power gain is maximised if both terms $\frac{R_{in} R_s}{(R_{in} + R_s)^2}$ and $\frac{R_L R_{out}}{(R_L + R_{out})^2}$ are maximised.

Consider the term $\frac{R_{in}R_s}{(R_{in} + R_s)^2}$ as an example. To find the input impedance that maximises

this term, we must determine its derivative with respect to R_{in} :

$$\Rightarrow \frac{d}{dR_{in}} \left(\frac{R_{in}R_s}{(R_{in} + R_s)^2} \right) = R_s \frac{d}{dR_{in}} \left(\frac{R_{in}}{(R_{in} + R_s)^2} \right) = R_s \frac{d}{dR_{in}} \left(R_{in} (R_{in} + R_s)^{-2} \right)$$

$$\Rightarrow R_s \frac{d}{dR_{in}} \left(R_{in} (R_{in} + R_s)^{-2} \right) = R_s \left[(R_{in} + R_s)^{-2} - 2R_{in} (R_{in} + R_s)^{-3} \right]$$

$$\Rightarrow R_s \left[(R_{in} + R_s)^{-2} - 2R_{in} (R_{in} + R_s)^{-3} \right] = R_s \left[\frac{1}{(R_{in} + R_s)^2} - \frac{2R_{in}}{(R_{in} + R_s)^3} \right]$$

$$\Rightarrow R_s \left[\frac{1}{(R_{in} + R_s)^2} - \frac{2R_{in}}{(R_{in} + R_s)^3} \right] = R_s \left[\frac{R_s - R_{in}}{(R_{in} + R_s)^3} \right].$$

It is easy to see that $\frac{d}{dR_{in}} \left(\frac{R_{in}R_s}{(R_{in} + R_s)^2} \right) = 0$ for $R_{in} = R_s$. This implies that the power gain is maximised for $R_{in} = R_s$.

In the same way, we can also show that the power gain is maximised for $R_{out} = R_L$.

We finally conclude that the transfer of power from source to load is maximised when $R_{in} = R_s$ and $R_{out} = R_L$.

The maximal value of the power gain is given by $A_p = A_v A_i \frac{(R_{in})^2}{(2R_{in})^2} \frac{(R_{out})^2}{(2R_{out})^2} = \frac{A_v A_i}{16}$.

The maximal power gain of an amplifier only depends on the OC voltage and SC current gains, and can indeed be very high when both A_v and A_i are high.

Before closing this chapter, it is important to mention that, from now on, we will almost exclusively focus our attention on the voltage amplifier model due to the high number of applications that require voltage amplifiers.

We will consider various circuits and find their unilateral voltage amplifier models, i.e. determine the expressions and values of their input resistances, output resistances, and OC voltage gains.

We will therefore assume by default that any circuit has a unilateral model. If this not the case, we will mention it and then determine the bilateral model of this circuit instead.